

Evaluation of heat transfer coefficient between a reacting steam-methane mixture and preheated Ni-based catalyst

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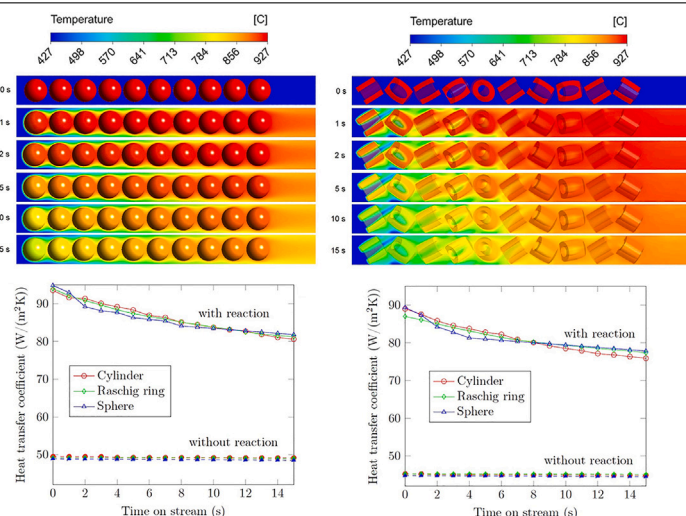
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HIGHLIGHTS

- 3-D unsteady CFD-model for steam methane reforming over a preheated catalyst.
- Two cases of heat transfer between catalyst and gas flow: with and without reaction.
- Increase in a steam-to-methane ratio leads to decrease in a heat transfer coefficient.
- Heat transfer coefficient for reacting flow is about twice higher than for non-reacting flow.

GRAPHICAL ABSTRACT



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ABSTRACT

This work is devoted to the investigation of a heat transfer inside a steam methane reformer filled with pre-heated Ni-based catalyst pellets of various shapes. Various particle shapes were tested: spheres, cylinders and Raschig ring under various inlet steam-to-methane ratios. The chemical kinetics model is based on experimentally derived reaction rate expressions taken from the literature. The distinguishing feature of the developed model is unsteady flow (transient) of reacting mixture over pre-heated Ni-based catalysts for the following initial parameters: inlet velocity – 0.5–1.7 m/s, inlet gas temperature – 400–450 °C, temperature of the preheated catalysts – 927 °C. This model allowed us to investigate the effect of endothermic chemical reactions on heat transfer between the catalysts and gas flow. Moreover, two cases were tested to understand the heat transfer in the reformer: the chemical reaction was switched on and switched off in the model. It was established that the heat transfer coefficient for a reacting flow ($\approx 90 \text{ W}/(\text{m}^2\text{K})$) is about twice higher than the heat transfer coefficient for a non-reacting flow.

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1. Introduction

An investigation of the steam methane reforming process in the 21st century has received a second wind. The emerging transition to carbon-free energy is associated with hydrogen by many experts [1,2]. About 95% of hydrogen in the World is produced by steam reforming of natural gas and other fossil hydrocarbons [3,4]. Moreover, the catalytic reforming processes, in particular the steam methane reforming process, are key hydrogen production technologies from hydrocarbons [5,6]. Therefore, steam methane reforming will play an important role as one of the main ways for hydrogen production in the nearest future.

Despite the abundance of studies on steam methane reforming (SMR), the number of articles in scientific journals on SMR is increasing from year to year [7–9]. This fact can be explained due to the complexity of physical and chemical processes inside reformers. The efficiency of the steam methane reforming process depends on many parameters. These parameters can be virtually divided into two groups: operational – temperature, pressure, inlet steam-to-methane ratio; reformer design – reformer geometry, type and shape of catalysis, residence time. To establish the relationship between these parameters and the efficiency of the steam methane reforming process, experimental investigation and numerical simulation are used [10–12]. However, an experiment is not always capable of characterizing the full effects of all these parameters [13,14].

Modern numerical methods such as CFD-modelling are used for deeper investigation of the phenomena in the steam methane reforming process [15]. Numerical simulation of the steam methane reforming process is a quite complicated task because in a methane reformer a lot of different processes run simultaneously: heat and mass transfer, diffusion, chemical reactions [16]. The increase in computing power and the development of software make the numerical study of the steam methane reforming process accessible to a wide range of researchers. Therefore, numerical models have become well-established tools for understanding and optimizing steam methane reformers [17,18]. A set of papers have been published on the CFD-modelling of lab-scale methane reformers and industrial-scale steam methane reformers [19–22].

In a study led by Cherif et al. [23] a steam-methane reforming reactions with methane combustion in an autothermal reformer were modelled. The authors used a finite-element approach to model the steam methane reforming reactions on the new catalyst arrangement. They concluded that patterned catalyst layers can enhance methane conversion rate, decrease the overall reactor length, thus making it more compact and stabilizing the catalyst by removing carbon molecules via oxidation reactions.

With the help of numerical calculation, it is relatively easy to compare the hydrogen productivity in different reactors. Thus, Lee and coauthors [24] decided to compare the hydrogen yield of 3 different technologies: a packed-bed reactor, membrane reactor and sorption-enhanced membrane reactor. The calculation was completed with the help of Matlab software. The separation of H_2 and CO_2 in a sorption-enhanced reactor led not only to higher H_2 yield but also to CO_2 capture, which makes the process significantly more eco-friendly.

One of the main tasks in the study of methane steam reforming is to determine the parameters of heat and mass transfer in the reaction space of the reactor [25]. The reaction rate has a significant effect on heat and mass transfer [26]. In this case, the reaction rate depends on temperature, pressure, mass fraction of reacting components, etc.

There are a set of papers on numerical investigation of heat and mass transfer realized in the commercial and open-source software [27]. Dixon [28] presented the results of 3-D simulations of a heterogeneous catalytic reaction of steam methane reforming in a fixed-bed reformer filled with spherical catalysts. Dixon used a realistic geometry of a reformer filled with 807 spherical particles. To simulate the chemical reactions he used the conventional Langmuir–Hinshelwood–Hougen–Watson kinetics approach. The simulations were performed via Ansys Fluent software. Dixon reported significant local variations in

temperature and species mass fraction from catalyst to catalyst, and in some cases from position to position on the same catalyst surface.

One of the important advantages of CFD modelling is getting visual results at every point in the computational domain. Wehinger et al. [29] presented the results of steam methane reforming numerical simulation performed via Star CCM+ software. The authors determined the effect of radiation on the variations of temperature and species mass fraction in the reformer. This paper presents the contours of temperature and hydrogen mass fraction in the reformer. Authors showed that at high temperatures the effect of radiation is significant but at moderate to low temperatures the effect of radiation is negligible.

Some authors investigated the species and temperature distribution in an unsteady process. Wang et al. [30] used Ansys Fluent to create a 2-D model of a steam methane reformer. The heat supply was traditionally organized through a reactor wall. The authors were interested in finding the most influential inlet boundary condition as well as in the development of the control scheme for hydrogen mass fraction on the reformer exit. Results showed that among the mass flow inlet, steam-to-carbon ratio and mixture inlet temperature, the mass flow is the most influential on the hydrogen production rate.

Karthik and Buwa [31] reported the results of a numerical investigation of steam methane reforming performed via Ansys Fluent. They presented the particle-resolved CFD simulations of multilayered packed beds containing 30 particles. Researchers considered five different particle shapes. The numerical investigation was conducted to determine the effect of particle shapes on pressure losses in a reformer, methane conversion and efficiency of the methane steam reforming process. A similar investigation was performed by the same authors [32] to understand the effect of catalyst shape on fluid flow and heat transfer in the steam methane reforming process in a fixed layer. The authors concluded that the efficiency of SMR is maximal for catalysts shapes with internal voids (holes).

From a practical point of view, CFD modelling of steam methane reforming is an important issue, particularly for the design of thermochemical exhaust heat recuperation systems [33,34]. This paper focuses on investigating the heat and mass transfer process for a moderately cold steam-methane mixture over a preheated Ni-based catalyst. This method of implementing the steam methane reforming process forms the basis of thermochemical regenerative heat recovery processes [35]. For such systems, calculating the heat transfer coefficient is an extremely important task.

From practical point of view, CFD-modelling of steam methane reforming is important issue, in particular, for designing of thermochemical exhaust heat recuperation systems. This paper is focused on investigation of heat and mass transfer process for a moderately cold steam-methane mixture over preheated Ni-based catalyst. Such way of implementation of the steam methane reforming process is a basis of thermochemical regenerative heat recovery process. For such systems, the calculation of heat transfer coefficient is extremely important task.

In the papers mentioned above, a steady-state numerical simulation of steam methane reforming was performed with a constant heat supply to the reactor walls. For the engineering tasks, it is important to know the heat transfer coefficient between the catalyst and reaction mixture. The steady-state process describes only part of the processes occurring in the reformer. In the actual process, endothermic reactions are an unsteady process and the unsteady process occurs always until equilibrium is reached. Therefore, the present study aims to determine the heat transfer coefficient between pre-heated porous Ni-based catalysts and reaction mixture in the unsteady steam methane reforming process. For this goal, we used a particle-resolved CFD model realized via Ansys Fluent for a wide range of operation conditions and constructive parameters. In this paper, the heat transfer coefficients were compared for two cases: Ni-based porous catalyst and porous particles without Ni-based active element.

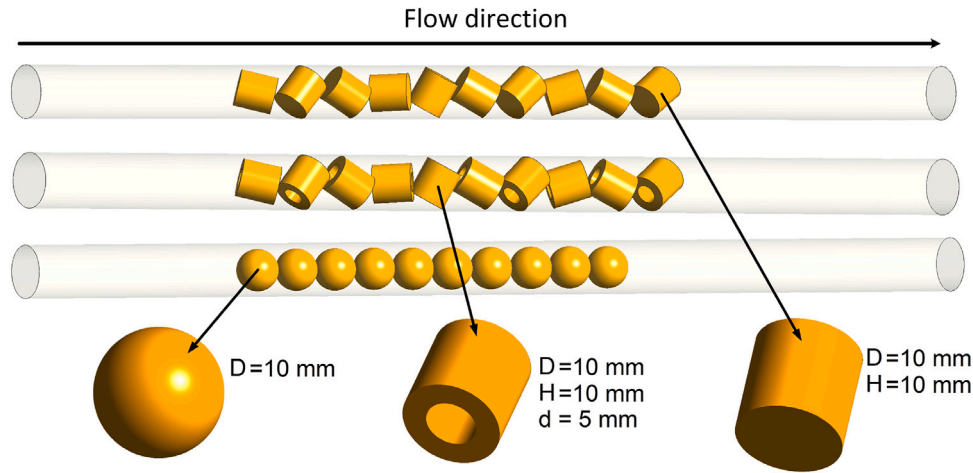


Fig. 1. The computational domain of a reformer filled with the catalysts: sphere, Rashcig ring, cylinder.

Table 1

Properties of catalyst.

Property	Units	Value
Density	kg/m ³	680
Heat capacity	J/(kg K)	600
Thermal conductivity	W/(m K)	5.8
Particle porosity	–	0.41
NiO	%	14.5
Surface area	m ² /kg _{cat}	4.5

2. CFD-modelling

To determine the heat transfer coefficient between catalyst and reaction mixture, three shapes of the catalyst were used: cylinder, Rashcig ring, and sphere. The properties of the Ni-based catalyst are plotted in Table 1.

2.1. Geometry and mesh

In this paper, the three most common catalyst shapes were analysed: cylinder, Rashcig ring and sphere. A computational domain consists of a tube filled with ten catalyst pellets of each shape. The computational domain for three catalyst shapes is presented in Fig. 1. A characteristic dimension for all shapes is 10 mm. An inner diameter of a Rashcig ring is 5 mm. An inner diameter of a tube is 14 mm. The flow direction is from left to right. Free space before and after the catalysts was added to get a developed flow before the catalysts and to exclude the reversed flows, respectively.

Based on the geometries from Fig. 1, the meshes were generated in Ansys Meshing module. Fig. 2 shows the mesh structure for a reformer filled with the cylindrical (a) and Rashcig ring (b) catalysts.

The mesh cells are tetrahedral of various sizes. Mesh refinement was used to correctly model the boundary layer in a flow region near the catalyst and tube walls. To get this refinement, *Inflation* procedure was used with the number of layers of 5. Moreover, the computational meshes were adapted to the dimensionless distance y^+ . An adaptation of the meshes were performed to meet the following condition – $y^+ < 50$. For a flow zone, y^+ is less than 25. However, in the area near the contact points between the catalyst particles and between the catalyst particle and tube wall, the y^+ is the following $25 < y^+ < 50$. The major goal of use of SWF – scalable wall function near the contact points is to force using a logarithmic law in combination with the standard turbulence calculation. This condition can be achieved by introducing a constraint in the determinations of y^+ to meet the following condition:

$$\bar{y}^+ = \text{MAX} (y^+; y_{lim}^+) \quad (1)$$

where y_{lim}^+ is maximum allowed value for each element of the mesh.

The mesh adaptations for y^+ from Eq. (1) were performed via the procedure *adapt* in Fluent solver.

The mesh independence studies were performed for all meshes. The average temperature of the gas mixture at the reformer outlet over a time step of 5 s was used as a reference parameter. Richardson extrapolation method was used for the analysis of the mesh Independence [36]. Fig. 3 shows the distribution of outlet temperature for the meshes with a different number of elements. Based on the mesh independence study, the following meshes were chosen: for spherical pellets – 4.8 mln elements, for cylindrical pellets – 7.8 mln elements, for Rashcig ring pellets – 8.2 mln elements.

2.2. Governing equations

The conservation, energy and momentum equations were solved in the developed model that was realized via Ansys Fluent. The catalyst particles were set as a porous zone, thus source term in a momentum equation was added for a porous medium domain. The steam methane reforming reactions were modelled for the transient regime (unsteady). The heat flux through the tube walls is zero (adiabatic walls).

The continuity and momentum equations for unsteady-state flow have the following form, respectively [37]:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0 \quad (2)$$

$$\frac{\partial}{\partial t} (\rho \vec{v}) + \nabla \cdot (\rho \vec{v} \vec{v}) = -\nabla \cdot p + \nabla \cdot (\bar{\tau}) + \rho \vec{g} + \vec{F} + S_i \quad (3)$$

where p – the static pressure; $\rho \vec{g}$ and $\vec{F}$ – the gravitational body force and external body forces, respectively; S_i – momentum source term for porous medium.

Stress tensor $\bar{\tau}$ in Eq. (3):

$$\bar{\tau} = \mu \left[(\nabla \vec{v} + \vec{v} \nabla) - \frac{2}{3} \nabla \cdot \vec{v} I \right] \quad (4)$$

where μ – molecular viscosity, I – unit tensor, and the second term on the right hand side is the effect of volume dilation.

The CFD modelling involves heat transfer. Therefore the viscosity was defined as a function of temperature by Sutherland's law via *Sutherland Law dialog box* (Two Coefficient Method) in Fluent solver.

A source term for porous medium S_i includes two terms: a viscous loss term, and an inertial loss term. In this paper, the porous medium of catalysts is considered as homogeneous. Therefore, the expression for S_i has the following form:

$$S_i = - \left(\frac{\mu}{\alpha} v_i + C_2 \frac{1}{2} \rho |v| v_i \right) \quad (5)$$

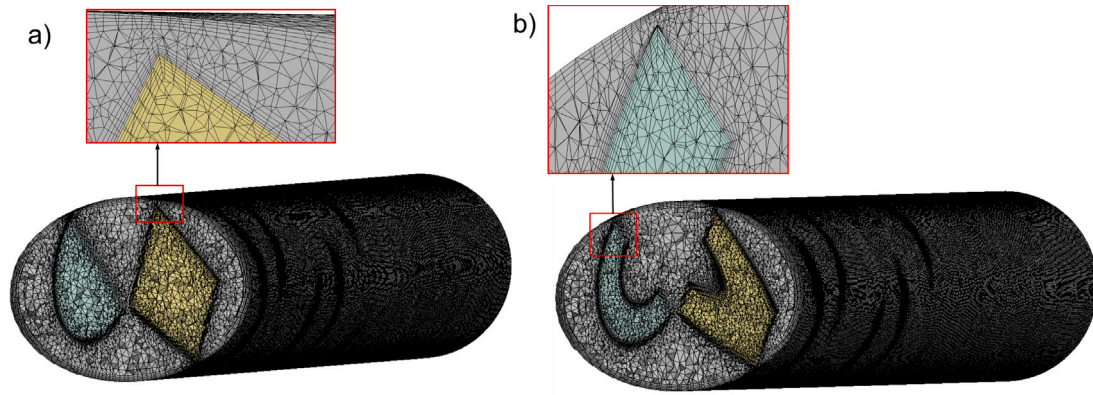


Fig. 2. The meshes of a reformer filled with the cylindrical (a) and Raschig ring (b) catalysts.

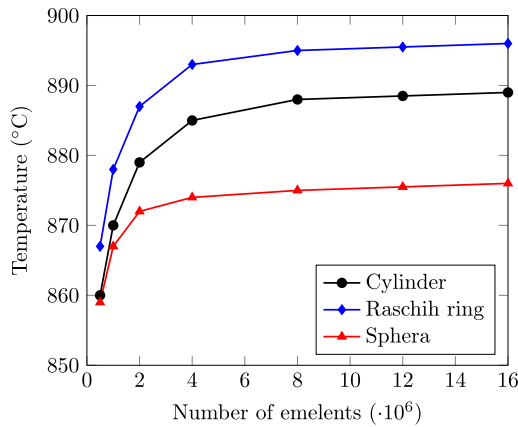


Fig. 3. Mesh independence study (reference parameter is outlet temperature).

where α – the permeability; C_2 – the inertial resistance factor; μ – the viscosity.

The coefficients α and C_2 were calculated (according to recommendations of Ansys Fluent Theory Guide) from the Ergun equation [38,39]:

$$\alpha = \frac{D_p^2}{150} \frac{\epsilon_{ip}^3}{(1 - \epsilon_{ip})^2} \quad (6)$$

and

$$C_2 = \frac{3.5}{D_p} \frac{(1 - \epsilon_{ip})}{\epsilon_{ip}^3}. \quad (7)$$

where D_p – the mean diameter of pore of the catalysts pellets, ϵ_{ip} – the in-particle porosity.

To solve RANS equation, RNG $k - \epsilon$ model is used [37].

The energy conservation equations for each reaction component in the reaction zone are expressed as:

$$\frac{\partial}{\partial t}(\rho H) + \nabla \cdot (\vec{u}(\rho + \rho \cdot h_f)) = \nabla \cdot [k_{eff} \cdot \nabla T - (\sum h_j \cdot \vec{J}_j) + \vec{\tau} \cdot \vec{v}] + S_f^h \quad (8)$$

where T – reaction mixture temperature; h_f – reaction mixture enthalpy; $\vec{J}_j$ – diffusion flux of j element; k_{eff} – conductivity; h_j – enthalpy of j element; S_f^h – gas enthalpy source term (include the source of energy due to chemical reaction).

The effective thermal conductivity in the porous medium, k_{eff} , is computed as the volume average of the fluid conductivity and the solid conductivity:

$$k_{eff} = \epsilon_{ip} k_f + (1 - \epsilon_{ip}) k_s \quad (9)$$

where k_f – fluid phase thermal conductivity (including the turbulent contribution, k_t); k_s – solid medium thermal conductivity

The fluid thermal conductivity k_f and the solid thermal conductivity k_s was computed via user-defined functions.

A mass transfer of j component of gas mixture can be described by following equation:

$$\frac{\partial}{\partial t}(\rho Y_j) + \nabla \cdot (\rho \cdot \vec{v} \cdot Y_j) = -\nabla \cdot \vec{J}_j + R_j \quad (10)$$

where R_j – reaction rate of formation or consumption of j th component; Y_j – mass fraction of j th component of gas mixture; $\vec{J}_j$ – diffusion flux of j component.

The diffusion flux of j th component can be expressed as:

$$\vec{J}_j = - \left(\rho \cdot D_{j,m} + \frac{\mu_t}{Sc_t} \nabla \right) Y_j - D_{T,j} \frac{\nabla T}{T} \quad (11)$$

where $D_{j,m}$ and $D_{T,j}$ – mass diffusion coefficient and thermal diffusion coefficient for j species, respectively; Sc_t and μ_t – turbulent Schmidt number and turbulent viscosity, respectively.

For the intra-particle volume, the transport process is determined by in-particle diffusion and chemical reactions. The structure of the porous particles is very complex because it consists of many interconnected pores with various forms, various diameters, etc. Xu and Froment proposed to use the following expression for an effective diffusion inside the catalyst particles for j th element [40]:

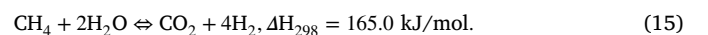
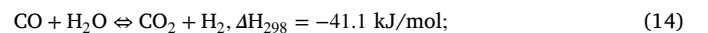
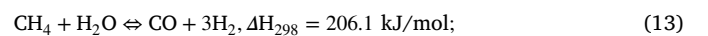
$$D_{eff,j} = \frac{\epsilon}{\tau} \bar{D}_j, \quad (12)$$

where τ – tortuosity factor; ϵ – porosity of catalyst particle; $\bar{D}_j$ – diffusion coefficient of j element.

A tortuosity factor in Eq. (12) can be varied from 1.5 to 8 for the different catalyst supports. There are no appropriate equations to calculate tortuosity; therefore, this parameter can be estimated experimentally. Wakao and Smith proposed to calculate tortuosity as $\tau = 1/\epsilon$ [41]. Various experimental investigations showed a good agreement with this expression. For example, tortuosity is from 2 to 2.5 for α -Al₂O support.

Temperature in the reaction space is less than 1000 °C and the effect of radiant heat transfer in the reaction flow is not taking into account according to paper [29].

The reaction rates in Eq. (10) were determined based on the chemical kinetics. A large amount of the kinetic models were developed for steam methane reforming. One of the most extensive studies of steam methane reforming over an industrial Ni-based catalyst was performed by Xu and Froment [42]. Intrinsic rate equations were obtained for the steam methane reforming reactions, accompanied by a water–gas shift reaction over a Ni-based catalyst:



Xu and Froment obtained the expression for the methane reforming rate:

$$r_{\text{CH}_4} = \left[\frac{k_1}{p_{\text{H}_2}^{2.5}} \left(p_{\text{CH}_4} \cdot p_{\text{H}_2\text{O}} - \frac{p_{\text{H}_2}^3 \cdot p_{\text{CO}}}{K_{e1}} \right) + \frac{k_3}{p_{\text{H}_2}^{3.5}} \left(p_{\text{CH}_4} \cdot p_{\text{H}_2\text{O}}^2 - \frac{p_{\text{H}_2}^4 \cdot p_{\text{CO}_2}}{K_{e3}} \right) \right] \cdot \frac{1}{Q_r} \quad (16)$$

In this expression, Q_r is denominator that can be determined as following:

$$Q_r = 1 + K_{\text{CO}} \cdot p_{\text{CO}} + K_{\text{H}_2} \cdot p_{\text{H}_2} + K_{\text{CH}_4} \cdot p_{\text{CH}_4} + \frac{K_{\text{H}_2\text{O}} \cdot p_{\text{H}_2\text{O}}}{p_{\text{H}_2}}, \quad (17)$$

Xu and Froment's model shows the 'non-Arrhenius' type behaviour of SMR. Also, they observed that the apparent activation energy in the power-law changes under closed circuit conditions. The authors determined all parameters for Eq. (16). All constants for Eqs. (16) and (17) were taken from Xu's paper [42].

2.3. Boundary conditions

In this paper, the steam methane reforming process occurs on a pre-heated Ni-based catalyst with the properties from Table 1. To understand the effect of the endothermic chemical reaction on heat transfer, two processes were considered. The first process is heat transfer between the catalysts and reaction mixture when chemical reactions are switched off (without chemical reaction, only heat transfer between catalyst and gas mixture). The second one is heat transfer between the catalysts and reaction mixture when the chemical reactions are active (with a chemical reaction based on Xu and Froment's kinetic model). The initial temperatures of the catalysts and reaction mixture were the same for all shapes of pellets. Catalyst temperature is notably higher than reaction mixture temperature. Therefore, heat transfer from the catalysts to the reaction mixture will be considerable even without the reactions [43]. In the process without chemical reactions, the temperature of a gas mixture is increasing and the temperature of the catalysts is decreasing over time. When chemical reactions occur, it is expected that the temperature of the catalysts will decrease faster than in the process without the chemical reactions. That is possible due to heat consumption in the course of the endothermic chemical reactions (13) and (15).

For each geometry from Fig. 1, two compositions of the gas mixture are analysed: steam-to-methane ratio (β) of 1 and 2. The velocities for all cases were the same.

For the transient models, time steps have a significant effect on the results. In this simulation, the Courant numbers were analysed to get $Cr < 1$. Therefore, the time step is set at 0.001 s. The total simulation time is 15 s. The time on stream of 15 s is chosen because it is enough to understand the effect of endothermic reactions on the heat transfer coefficient. For the design of thermochemical exhaust heat recuperation systems, a time on steam greater than 15 s can be chosen. As shown below, after 10 s, the heat transfer coefficient monotonically decreases with time.

All results obtained with help of this CFD model were calculated with second-order upwind interpolation for the convection terms; all diffusion terms used second-order discretization. The convergence criterion for momentum, continuity and species are defined as 10^{-3} , for energy less than 10^{-6} . The hybrid initialization is chosen to initialize the solution. To enhance the stability of the calculations, the under-relaxation factors for momentum were reduced to 0.55, and the scalars were also reduced to 0.95.

The boundary conditions are determined as follows:

- reformer inlet:

$$x = 0; T_g = T_{g(in)}; C_i = C_{i(in)}. \quad (18)$$

- reformer outlet:

$$x = L; \frac{\partial C_i}{\partial x} = 0. \quad (19)$$

where C_i – mole concentration of i element.

The main assumptions are taken in this model:

- radiative heat transfer is neglected;
- gas flow is incompressible;
- buoyancy and viscous heating effects are neglected;
- catalyst particle porosity is isotropic;

3. Results

The CFD modelling was conducted with the use of ANSYS Fluent 19 R2 on a PC with a 16-core Intel Core i9 X-series with 96 GB RAM. To simulate the chemical kinetic according to the expressions (Eqs. (16) (17)), a user-defined function (UDF) was prepared and compiled in C++. *DEFINE_VR_RATE* was used to specify a custom volumetric reaction rate for multiple reactions in steam methane reforming. During ANSYS FLUENT execution, *DEFINE_VR_RATE* is called for every reaction in every single cell [44,45].

One of the main advantages of CFD modelling is obtaining visible results for the different parameters. Fig. 4 shows the temperature contours of the stream with an active chemical reaction in the steam methane reformer filled with sphere pellets (a) and Rashcig ring pellets (b) at a steam-to-methane ratio of 1 for various time steps. The initial temperatures are 427 °C for a gas mixture and 927 °C, inlet velocity is 1.5 m/s. The contours were fixed at the following time steps: 0, 1, 2, 5, 10, 15 s.

Fig. 4 shows that during the flow over the preheated catalysts the temperature of a gas mixture is increasing and the temperature of the catalysts is decreasing. At the initial time step (0 s) there is no heat transfer between the catalysts and gas mixture. For the next time steps, the temperature of the catalysts is decreasing due to the heating of a gas mixture and chemical endothermic reactions. The maximum cooling of the catalysts is observed on the first catalysts pellets.

The pellets in the reformer filled with Rashcig rings are cooling faster than the spherical pellets because the heat transfer surface in the reformer filled with Rashcig rings is higher than in the reformer filled with spheres [46]. To understand the mass transfer in the reformer, Fig. 5 shows the contours of hydrogen and methane mass fractions for the different time steps. The contours in Fig. 5 were obtained for the same initial conditions as the contours in Fig. 4.

Fig. 5 shows that for the first time steps (1–5 s), the methane conversion is higher than for the last time steps (10–15 s) because the reaction rate depends on temperature. The temperature of the first catalysts is getting quite low and therefore the reaction rate of methane reforming is very small. As it can be seen, the very low methane conversion can be observed in the first three pellets. However, for the last pellets, the methane conversion is moderately high because the temperature of these pellets is quite high. The same behaviour is observed on the hydrogen mass fraction contours. The maximum mass fraction of H_2 is located inside the porous pellets as velocity and diffusion in the porous medium are quite low and temperature is high.

As mentioned above, to understand the effect of the endothermic reaction on the heat transfer between the catalyst pellet and gas flow, two cases were compared: flow with the steam methane reforming reactions and flow without the steam methane reactions. For the case without reactions, the chemical mechanism was switched off. Fig. 6 shows the temperature contours at a different operating time in the steam methane reformer filled with spherical pellets for two cases: (a) with the chemical reactions and (b) without the chemical reactions. Fig. 6 depicts that as the flow time increases the surface temperature of catalyst pellets decreases slower for the case without the chemical reactions. On the other side, the temperature of gas flow after 15 s is higher for the case without the chemical reaction in comparison with

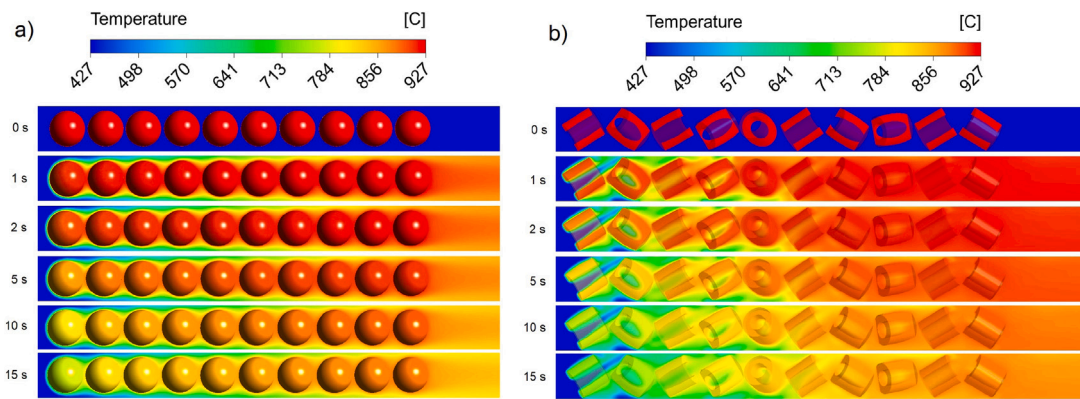


Fig. 4. Temperature contours as a function time on stream in the steam methane reformer filled with spherical pellets (a) and Rashcig ring pellets (b) at steam-to-methane ratio of 1.

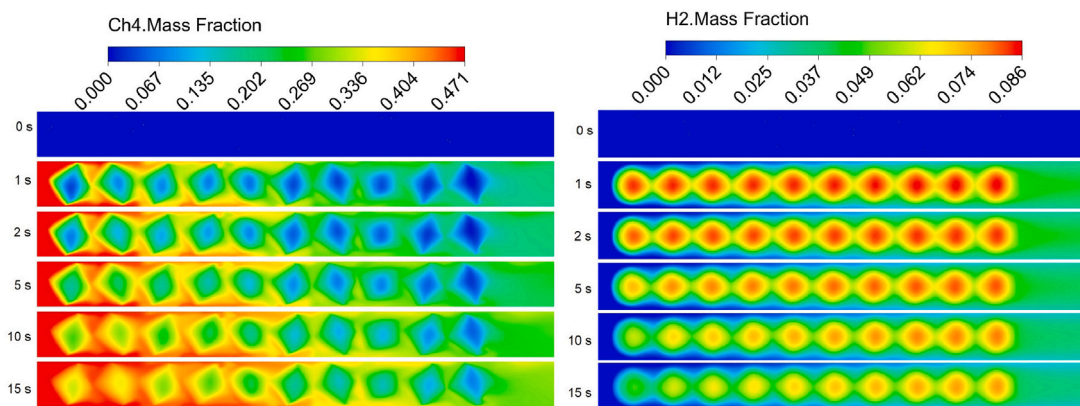


Fig. 5. Contours of CH_4 and H_2 mass fraction as a function time on stream in the steam methane reformer filled with spherical pellets and Rashcig ring pellets (b) at steam-to-methane ratio of 1.

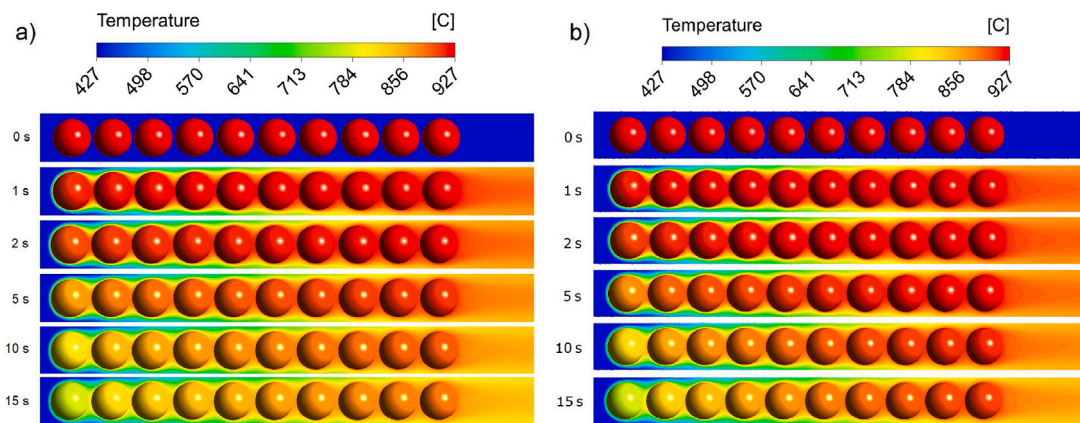


Fig. 6. Temperature contours at different operating time in the steam methane reformer filled with spherical pellets for two cases: (a) with the chemical reactions and (b) without the chemical reactions.

the case with the chemical reactions. Thus, based on the temperature contours in Fig. 6 it can be concluded that the endothermic chemical reaction leads to an increase in the heat transfer between the catalyst and gas flow.

To quantify the effect of chemical reactions on the temperature distribution, the temperature of pellets was compared for two cases: flow with active steam methane reforming reactions and flow without the steam methane reforming reactions. Fig. 7 shows the volume average temperature for the different pellets in the reformer at various flow time.

Fig. 7 shows the average volume temperature of the 1st, 5th and 10th catalyst particles (in Fig. 1, the 1st particle is on the left, the 10th particle is on the right) over time at a steam-to-methane ratio of 1. Solid lines – reacting steam-methane mixture, dashed line – non-reacting mixture. Fig. 7a shows the temperature distribution for cylindrical particles, Fig. 7b shows the temperature distribution for Rashcig rings. It can be seen that if the reaction does not proceed, the dependence of the catalyst temperature on time is nearly linear for cylinders and Rashcig rings. Meanwhile, at the beginning of the flow, the Rashcig rings cool faster than the cylinders. This is because

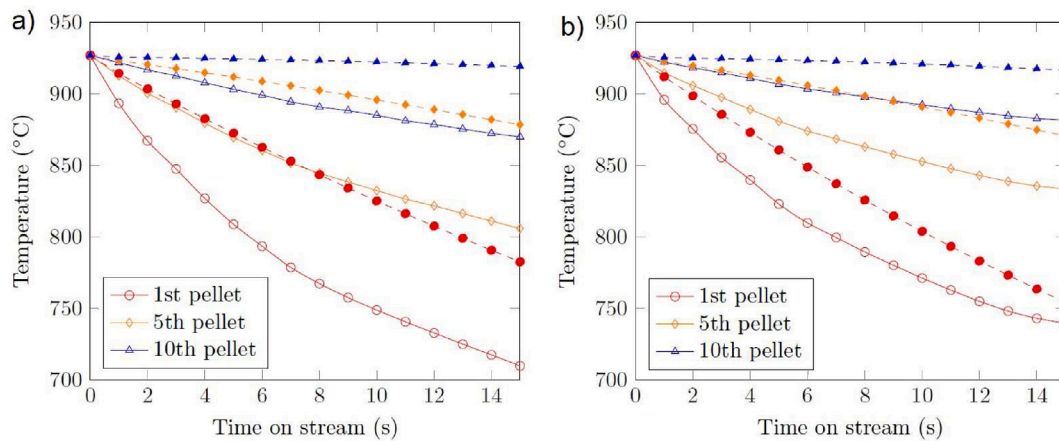


Fig. 7. Catalyst volume average temperature of 1st, 5th and 10th pellets during time on stream of 15 s for steam-to-methane ratio of 1.0: (a) cylindrical pellets; (b) Raschig ring pellets. Solid line – reacting flow (with reaction); dashed line – non-reacting flow (without reaction).

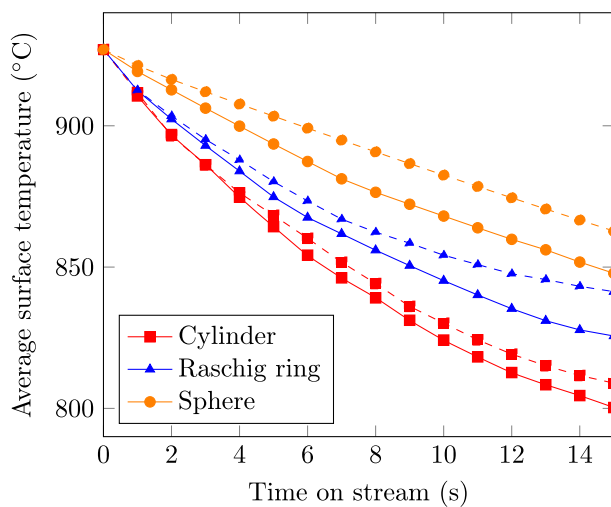


Fig. 8. Volume average temperature of catalyst pellets during time on stream for steam-to-methane ratio of 1.0 (solid lines) and 2.0 (dashed lines).

the heat exchange surface area in Raschig rings is higher, and the total heat capacity of the particle is smaller. When a chemical reaction occurs, the dependence of particle temperature on the flow time is non-linear [47]. At first, the temperature drops sharply, and then the change becomes close to linear. The reason for this behaviour is that the heat of the catalyst is spent not only on heating the gas mixture but also on the endothermic chemical reactions. The reaction rate decreases when the temperature decreases and the concentration of methane decreases. Therefore, after 10 s, the chemical reaction on the first particles does not have a significant effect on heat transfer, and the change in the catalyst temperature occurs mainly due to the heating of the gas mixture.

To understand the effect of a steam-to-methane ratio on heat transfer in the steam methane reformer, two steam-to-methane ratios of 1.0 and 2.0 were tested at the same temperatures ($T_{cat} = 927$ °C and $T_{gas,in} = 427$ °C) and velocity (1.5 m/s). Fig. 8 shows the average volume temperature of catalyst pellets during the time on stream for the steam-to-methane ratio of 1.0 (solid lines) and 2.0 (dashed lines).

Fig. 8 makes it clear that an increase in a steam-to-methane ratio leads to an increase in average catalyst temperature. The main reason for this effect is decreasing in partial pressure of methane which leads to a decrease in the rate of reforming reaction and as a consequence to a decrease in heat consumption by the endothermic process. Therefore,

it can be concluded that an increase in the steam-to-methane ratio leads to a decrease in the total heat transfer coefficient.

To determine the heat transfer coefficient, the classical Newton's law for a temperature-independent heat transfer coefficient is utilized [48]:

$$\dot{Q} = hF (T_{cat}(t) - T_{gas}), \quad (20)$$

where $\dot{Q}$ – the rate of heat transfer out of the catalyst due to gas flow heating and endothermic chemical reaction; h – the heat transfer coefficient (assumed independent of T and averaged over the surface); F – the heat transfer surface area; T – the temperature of the catalyst's surface; T_{gas} – the temperature of the gas mixture.

Obviously, the heat transfer for the reacting flow includes the term for heat consumption during endothermic reactions and the mixture heating. Therefore, further we will consider two heat transfer coefficients: heat transfer coefficient for non-reacting flow ($h_{n/r}$) and heat transfer coefficient for reacting flow (h_r). Fig. 9 shows the average heat transfer coefficients in the steam methane reformer filled with spheres, cylinders and Raschig rings for a steam-to-methane ratio of 1.0 (a) and 2.0 (b).

Fig. 9 depicts the interesting findings. The heat transfer coefficients for non-reacting flow are almost similar for all catalysts shapes: about 49.8 W/(m² K) for a steam-to-methane ratio of 1.0 and about 46.2 W/(m² K) for a steam-to-methane ratio of 2.0. The dependence of the heat transfer coefficient on time on stream shows that the very steep reduction is observed for both mixtures with steam-to-methane reactions 1 and 2. The heat transfer coefficient reduction could be much more pronounced when the last (10th) catalyst starts active cooling. The process was running for just 15 s so the outlet mixture temperature varied insignificantly. After the 10th catalyst starts cooling, the outlet mixture temperature will begin to decrease fast, and so will the heat transfer coefficient. This case makes it clear that in the reformer without chemical reaction the heat transfer coefficient depends in the early stages of the process mainly on gas velocity and gas composition.

In addition, Fig. 9 shows that switching on the endothermic chemical reactions leads to a significant increase in the heat transfer coefficient. This increase is taking place due to heat consumption by the endothermic reactions. The comparison of the heat transfer coefficients for two steam-to-methane ratios shows that a heat transfer coefficient for $\beta = 1.0$ is higher than for $\beta = 2.0$. This fact can be explained by a reduction in the reaction rate due to a decrease in methane concentration. Moreover, Fig. 9 depicts that the heat transfer coefficient for a reacting flow is constantly decreasing during the time on stream. This happens due to a reduction in temperature and methane partial pressure in a reacting flow that leads to a lower reaction rate which in turn leads to a lower heat consumption by endothermic reactions.

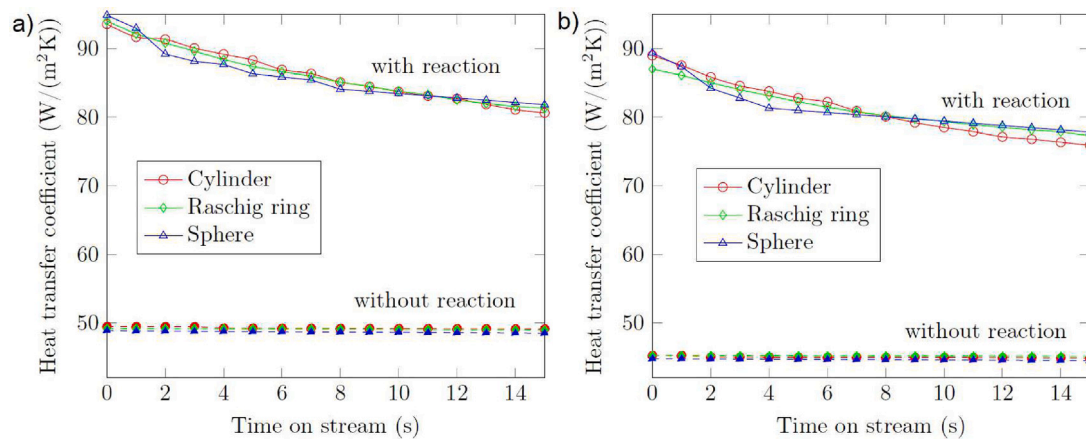


Fig. 9. Average heat transfer coefficient in the steam methane reformer for a steam-to-methane ratio of 1.0 (a) and 2.0 (b).

Fig. 9 makes it clear that for increasing the efficiency of the steam reforming it is important to increase the heat transfer surface because the shape of catalyst pellets does not play a significant role in the heat transfer.

4. Conclusion

This work presents the results of a numerical investigation of the heat transfer coefficient between the catalysts and reaction mixture for the steam reforming process. The commercial software Ansys Fluent was used to calculate the heat transfer for 3D geometry. A global mechanism of Langmuir–Hinshelwood–Hougen–Watson kinetics was taken from the fundamental paper presented by Xu and Froment. A reformer filled with the pre-heated catalysts of three shapes (sphere, cylinder, Raschig ring) were tested. To understand the effect of the chemical endothermic process on heat transfer coefficient, two cases were tested: with chemical reaction and without chemical reaction. The results of CFD simulations revealed the preheated catalyst pellets are cooling faster when the chemical reaction is occurring in a comparison with the case without a chemical reaction. Thus, the endothermic chemical reaction leads to an increase in the heat transfer between the catalyst and gas flow. It was established that the catalysts in the reformer with the chemical reactions filled with Raschig rings are cooling faster than the spherical and cylindrical catalysts because the heat transfer surface in the reformer filled with Raschig rings is higher than in the reformer filled with spheres and cylinders. The results showed that the heat transfer coefficient does not considerably depend on the catalyst shapes. For all three shapes, the dependences of the change in heat transfer coefficient as a function of time on stream are almost similar. The different steam-to-methane ratios (1.0 and 2.0) were also tested. With an increase in a steam-to-methane ratio, the partial pressure of methane is decreasing therefore the rate of reforming reaction is also decreasing. The results showed the heat transfer coefficient for a steam-to-methane ratio of 2.0 is ≈ 5 W/(m² K) less than for a steam-to-methane ratio of 1.0. The results showed that the average heat transfer coefficient for a reacting flow (≈ 90 W/(m² K)) is about twice higher than the heat transfer coefficient for a non-reacting flow (≈ 46 W/(m² K)) for all investigated shapes.

CRediT authorship contribution statement

Dmitry Pashchenko: Methodology, Writing, Conceptualization.
Igor Karpilov: Modelling, Visualization, Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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References

- [1] Arun Majumdar, John M Deutch, Ravi S Prasher, Thomas P Griffin, A framework for a hydrogen economy, *Joule* 5 (8) (2021) 1905–1908.
- [2] Nazim Z. Muradov, T. Nejat Veziroğlu, “Green” path from fossil-based to hydrogen economy: an overview of carbon-neutral technologies, *Int. J. Hydrogen Energy* 33 (23) (2008) 6804–6839.
- [3] Ibrahim Dincer, Canan Acar, Review and evaluation of hydrogen production methods for better sustainability, *Int. J. Hydrogen Energy* 40 (34) (2015) 11094–11111.
- [4] Ibrahim Dincer, Calin Zamfirescu, *Sustainable Hydrogen Production*, Elsevier, 2016.
- [5] Trevor L. LeValley, Anthony R. Richard, Maohong Fan, The progress in water gas shift and steam reforming hydrogen production technologies—a review, *Int. J. Hydrogen Energy* 39 (30) (2014) 16983–17000.
- [6] Jens R. Rostrup-Nielsen, Thomas Rostrup-Nielsen, Large-scale hydrogen production, *Cattech* 6 (4) (2002) 150–159.
- [7] Igor Karpilov, Dmitry Pashchenko, Effect of methane reforming before combustion on emission and calorimetric characteristics of its combustion process, *Int. J. Hydrogen Energy* 46 (76) (2021) 38106–38118.
- [8] Zhihong Wu, Jingyu Wang, Bingjing Xie, Jian Yang, Qiuwang Wang, Methane steam reforming with axial variable diameter particle structures in grille-sphere composite packed bed: A numerical study of hydrogen production performance, *Energy Convers. Manage.* 240 (2021) 114163.
- [9] Dmitry Pashchenko, Ravil Mustafin, Anna Mustafina, Steam methane reforming in a microchannel reformer: Experiment, CFD-modelling and numerical study, *Energy* 237 (2021) 121624.
- [10] Yury Voloshchuk, Andreas Richter, Reduced order modeling and large-scale validation for non-catalytic partial oxidation of natural gas, *Chem. Eng. Sci.* (2022) 117620.
- [11] Yi Ran Lu, Dmitry Pashchenko, Petr A. Nikrityuk, A new semiempirical model for the heat and mass transfer inside a spherical catalyst in a stream of hot CH₄/H₂O gases, *Chem. Eng. Sci.* 238 (2021) 116565.
- [12] Philipp Wachter, Christian Gaber, Martin Demuth, Christoph Hochenauer, Towards a recuperative, stationary operated thermochemical reformer: Experimental investigations on the methane conversion and waste heat recovery, *Appl. Therm. Eng.* 183 (2021) 116121.
- [13] Hessamodin Nourbakhsh, Javad Rahbar Shahrouzi, Hadi Ebrahimi, Akbar Zamanian, Mohammad Reza Jafari Nasr, Experimental and numerical study of syngas production during premixed and ultra-rich partial oxidation of methane in a porous reactor, *Int. J. Hydrogen Energy* 44 (60) (2019) 31757–31771.
- [14] Philipp Wachter, Christian Gaber, Martin Demuth, Christoph Hochenauer, Experimental investigation of tri-reforming on a stationary, recuperative TCR-reformer applied to an oxy-fuel combustion of natural gas, using a Ni-catalyst, *Energy* 212 (2020) 118719.

- [15] Behnam Partopour, Anthony G. Dixon, An integrated workflow for resolved-particle packed bed models with complex particle shapes, *Powder Technol.* 322 (2017) 258–272.
- [16] Moyu Wang, Panayotis Dimopoulos Eggenschwiler, Davide Ferri, Oliver Kröcher, Experimental and modeling-based analysis of reaction pathways on catalysts for natural gas engines under periodic lean/rich oscillations, *Chem. Eng. J.* 430 (2022) 132848.
- [17] Majid Siavashi, Farzad Hosseini, Hamid Reza Talesh Bahrami, A new design with preheating and layered porous ceramic for hydrogen production through methane steam reforming process, *Energy* 231 (2021) 120952.
- [18] Ali Cherif, Rachid Nebbali, Fatih Sen, John W. Sheffield, Nimeti Doner, Lyes Nasser, Modeling and simulation of steam methane reforming and methane combustion over continuous and segmented catalyst beds in autothermal reactor, *Int. J. Hydrogen Energy* (2022).
- [19] Mukesh Upadhyay, Hyunjun Lee, Ayeon Kim, Sang-hun Lee, Hankwon Lim, CFD simulation of methane steam reforming in a membrane reactor: Performance characteristics over range of operating window, *Int. J. Hydrogen Energy* 46 (59) (2021) 30402–30411.
- [20] Mustafa Tutar, Cihat Emre Üstün, Jose Miguel Campillo-Robles, Raquel Fuente, Silvia Cibrián, Ignacio Arzuza, Arturo Fernández, Gabriel A. López, Optimized CFD modelling and validation of radiation section of an industrial top-fired steam methane reforming furnace, *Comput. Chem. Eng.* 155 (2021) 107504.
- [21] Iman Rahimipetroudi, Jae Seon Shin, Kashif Rashid, Je Bok Yang, Sang Keun Dong, Development and CFD analysis for determining the optimal operating conditions of 250 kg/day hydrogen generation for an on-site hydrogen refueling station (HRS) using steam methane reforming, *Int. J. Hydrogen Energy* 46 (71) (2021) 35057–35076.
- [22] Rached Ben-Mansour, MD Azazul Haque, Aadesh Harale, Stephen N. Paglieri, Firas S. Alrashid, Mohammad Raghieb Shakeel, Esmail MA Mokheimer, Mohamed A. Habib, Comprehensive parametric investigation of methane reforming and hydrogen separation using a CFD model, *Energy Convers. Manage.* 249 (2021) 114838.
- [23] Ali Cherif, Rachid Nebbali, John William Sheffield, Nimeti Doner, Fatih Sen, Numerical investigation of hydrogen production via autothermal reforming of steam and methane over Ni/Al₂O₃ and Pt/Al₂O₃ patterned catalytic layers, *Int. J. Hydrogen Energy* 46 (75) (2021) 37521–37532.
- [24] Hyunjun Lee, Ayeon Kim, Boreum Lee, Hankwon Lim, Comparative numerical analysis for an efficient hydrogen production via a steam methane reforming with a packed-bed reactor, a membrane reactor, and a sorption-enhanced membrane reactor, *Energy Convers. Manage.* 213 (2020) 112839.
- [25] Ali Cherif, Rachid Nebbali, Chul-Jin Lee, Design and multiobjective optimization of membrane steam methane reformer: A computational fluid dynamic analysis, *Int. J. Energy Res.* (2022).
- [26] Philipp Wachter, Philipp Hödl, Juraj Raic, Christian Gaber, Martin Demuth, Christoph Hochenauer, Towards thermochemical recuperation applying combined steam reforming and partial oxidation of methane: Thermodynamic and experimental considerations, *Energy Convers. Manage.* 251 (2022) 114927.
- [27] M.N. Nikitin, D. Pashchenko, Statistical method of deducting activation energies for the steam methane reforming reactions, *J. Comput. Biophys. Chem.* (2022) 1–12.
- [28] Anthony G. Dixon, Local transport and reaction rates in a fixed bed reactor tube: Endothermic steam methane reforming, *Chem. Eng. Sci.* 168 (2017) 156–177.
- [29] Gregor D. Wehinger, Steffen Flaischlen, Computational fluid dynamics modeling of radiation in a steam methane reforming fixed-bed reactor, *Ind. Eng. Chem. Res.* 58 (31) (2019) 14410–14423.
- [30] Jingyu Wang, Shangshang Wei, Qiuwang Wang, Bengt Sundén, Transient numerical modeling and model predictive control of an industrial-scale steam methane reforming reactor, *Int. J. Hydrogen Energy* 46 (29) (2021) 15241–15256.
- [31] G.M. Karthik, Vivek V. Buwa, Particle-resolved simulations of methane steam reforming in multilayered packed beds, *AIChE J.* 64 (11) (2018) 4162–4176.
- [32] G.M. Karthik, Vivek V. Buwa, Effect of particle shape on fluid flow and heat transfer for methane steam reforming reactions in a packed bed, *AIChE J.* 63 (1) (2017) 366–377.
- [33] Dmitry Pashchenko, Ravil Mustafin, Ammonia decomposition in the thermochemical waste-heat recuperation systems: A view from low and high heating value, *Energy Convers. Manage.* 251 (2022) 114959.
- [34] Dmitry Pashchenko, Ravil Mustafin, Igor Karpilov, Efficiency of chemically recuperated gas turbine fired with methane: Effect of operating parameters, *Appl. Therm. Eng.* 212 (2022) 118578.
- [35] Hisashi Kobayashi, Thermochemical regenerative heat recovery process, 2000, US Patent 6, 113, 874.
- [36] Andrés Meana-Fernández, Jesus Manuel Fernandez Oro, Katia María Argüelles Díaz, Mónica Galdo-Vega, Sandra Velarde-Suárez, Application of richardson extrapolation method to the CFD simulation of vertical-axis wind turbines and analysis of the flow field, *Eng. Appl. Comput. Fluid Mech.* 13 (1) (2019) 359–376.
- [37] Cx K. Batchelor, G.K. Batchelor, *An Introduction to Fluid Dynamics*, Cambridge University Press, 2000.
- [38] ANSYS FLUENT Theory Guide, Ansys fluent theory guide. release 18.2, 15317 (november), 2013.
- [39] Dmitry Pashchenko, Igor Karpilov, Ravil Mustafin, Numerical calculation with experimental validation of pressure drop in a fixed-bed reactor filled with the porous elements, *AIChE J.* 66 (5) (2020) e16937.
- [40] Jianguo Xu, Gilbert F. Froment, Methane steam reforming: II. Diffusional limitations and reactor simulation, *AIChE J.* 35 (1) (1989) 97–103.
- [41] Noriaki Wakao, J.M. Smith, Diffusion in catalyst pellets, *Chem. Eng. Sci.* 17 (11) (1962) 825–834.
- [42] Jianguo Xu, Gilbert F. Froment, Methane steam reforming, methanation and water-gas shift: 1. Intrinsic kinetics, *AIChE J.* 35 (1989) 88–96.
- [43] Florent Minette, Luis Calamote de Almeida, Jonathan Feinstein, Juray De Wilde, Structured ZoneFlow™-bayonet steam reforming reactor for reduced firing and steam export: Pressure drop and heat transfer modelling and evaluation of the reactor performance, *Chem. Eng. J. Adv.* 10 (2022) 100258.
- [44] Anh Tran, Andres Aguirre, Helen Durand, Marquis Crose, Panagiotis D. Christofides, CFD modeling of a industrial-scale steam methane reforming furnace, *Chem. Eng. Sci.* 171 (2017) 576–598.
- [45] Dmitry Pashchenko, Anton Eremin, Heat flow inside a catalyst particle for steam methane reforming: CFD-modeling and analytical solution, *Int. J. Heat Mass Transfer* 165 (2021) 120617.
- [46] G.M. Karthik, Vivek V. Buwa, Effect of particle shape on catalyst deactivation using particle-resolved CFD simulations, *Chem. Eng. J.* 377 (2019) 120164.
- [47] Jinwon Yun, Sangseok Yu, Transient behavior of 5 kW class shell-and-tube methane steam reformer with intermediate temperature heat source, *Int. J. Heat Mass Transfer* 134 (2019) 600–609.
- [48] Yueqiang Ma, Quan Gan, Yanjun Zhang, Xuebing Yuan, Experimental study of heat transfer between fluid flowing through fracture surface with tortuous seepage path, *Renew. Energy* 188 (2022) 81–95.